

AETHERIS

A Collateralized Monetary Protocol

AETHERIS Protocol — V2 Whitepaper

AUSD Stablecoin · Vaults · Oracle Infrastructure · Protocol Fees · Treasury · AETR ·
Staking

Current deployment Ethereum Sepolia Testnet	Protocol assets AUSD · AETR
Architecture Collateralized Vault system	Version AETHERIS V2

This document describes the architecture and currently deployed functionality of AETHERIS V2. Where functionality is planned rather than deployed, it is explicitly identified as a future direction.

Contents

- 1. Abstract
- 2. Vision and Design Principles
- 3. Protocol Architecture
- 4. AUD Stablecoin
- 5. Vault and Collateral
- 6. Borrowing and Repayment
- 7. Liquidation Engine
- 8. Price Oracle
- 9. Protocol Fee Router
- 10. Treasury and Cold-Wallet Architecture
- 11. AETR Token
- 12. AETR Tokenomics
- 13. Staking
- 14. Security Architecture
- 15. Risk Management
- 16. Ethereum Sepolia Testnet
- 17. Smart Contract Addresses
- 18. Roadmap
- 19. Disclaimer

1. Abstract

AETHERIS is a collateralized monetary protocol designed around the AUSD stablecoin, user Vaults, on-chain risk management, price-oracle infrastructure, protocol fee routing, Treasury reserves, AETR token economics and staking.

The protocol separates major responsibilities across dedicated smart contracts. The Vault manages collateralized debt positions, AUSD provides the monetary accounting layer, the PriceOracle provides market data, the ProtocolFeeRouter coordinates protocol fee accounting, Treasury provides reserve infrastructure, and AETR and Staking provide the protocol's token and staking layers.

AETHERIS V2 is currently deployed and tested on Ethereum Sepolia. Testnet assets and balances should not be treated as production monetary reserves or financial guarantees.

2. Vision and Design Principles

The AETHERIS design is centered on transparent on-chain accounting and separation of protocol responsibilities.

- **Collateralization:** AUSD borrowing is associated with collateralized Vault positions.
- **On-chain accounting:** Core balances, debt, collateral, supply and protocol parameters are represented by smart contracts.
- **Risk controls:** Borrowing and liquidation behavior is governed by explicit contract parameters.
- **Oracle awareness:** Price data includes update information and freshness controls.
- **Reserve separation:** Treasury reserves are separated conceptually from active Vault collateral.
- **Operational security:** Administrative capabilities are restricted through contract-level authorization.

3. Protocol Architecture

AETHERIS V2 uses a modular contract architecture. Each major subsystem has a defined responsibility.

Component	Responsibility
Vault	Collateral, debt, borrowing, repayment, withdrawals and liquidation.
AUSD	Stablecoin supply, balances, minting and burning through access control.
PriceOracle	ETH, BTC and AETR price data and freshness information.
ProtocolFeeRouter	Protocol fee calculation and allocation accounting.
Treasury	Protocol reserves, token balances and designated cold-wallet infrastructure.
AETRToken	AETR supply, burn and controlled future-emission functionality.
Staking	AETR staking balances, deposits and withdrawals.

4. AUSD Stablecoin

AUSD is the primary stablecoin asset of the AETHERIS monetary layer. The deployed AUSD contract uses role-based authorization for privileged monetary operations.

Core properties

- ERC-20 compatible token interface.
- 18 decimal places in the deployed implementation.
- Role-controlled minting.
- Role-controlled burning.
- Administrative controls protected through access control.

AUSD lifecycle

A typical user lifecycle begins with collateral being deposited into the Vault. Subject to the Vault's risk parameters, the user can borrow AUSD. When the user repays AUSD debt, the outstanding Vault debt is reduced.

The deployed smart contract remains the source of truth for AUSD supply, balances, roles and monetary execution.

5. Vault and Collateral

The Vault is the principal collateralized debt engine. The deployed V2 Vault tracks ETH collateral and AUSD debt on a per-user basis.

User position

- ETH collateral deposited by the user.
- AUSD debt associated with the user.
- Collateral valuation based on the configured PriceOracle.
- Maximum borrowing capacity determined by Vault parameters.
- Withdrawal functionality subject to the deployed contract's checks.

The V2 Vault exposes parameters including maximum LTV, liquidation threshold, liquidation bonus, maximum liquidation close factor and borrowing fee.

6. Borrowing and Repayment

Borrowing

A user can request AUSD borrowing through the Vault. The Vault evaluates the user's position against its configured borrowing limits before executing the transaction.

Repayment

A user can repay AUD against the Vault debt. Successful repayment reduces the outstanding debt recorded for the user and can improve the health of the collateralized position.

Borrowing capacity, debt accounting and transaction execution are enforced by the Vault contract rather than by the website UI.

7. Liquidation Engine

The liquidation engine protects the collateralized debt system when positions no longer satisfy the configured risk parameters.

Relevant V2 parameters

- LIQUIDATION_THRESHOLD_BPS
- LIQUIDATION_BONUS_BPS
- MAX_LIQUIDATION_CLOSE_FACTOR_BPS
- MAX_LTV_BPS

The V2 Vault exposes an `isHealthy` function and a `liquidate` function. These mechanisms allow the deployed contract to determine position health and process eligible liquidations.

Liquidation parameters are contract-controlled. The website is an interface and monitoring layer and does not replace the smart contract's enforcement.

8. Price Oracle

The PriceOracle contract supplies market-price information used by the protocol. The deployed V2 implementation exposes ETH, BTC and AETR price data together with timestamp information.

Supported price feeds

- ETH price
- BTC price
- AETR price

Freshness

The oracle exposes a configurable maximum price age and functions for evaluating whether price information remains fresh.

Oracle freshness is a critical risk consideration because stale collateral prices can affect borrowing and liquidation decisions.

9. Protocol Fee Router

The ProtocolFeeRouter is the fee accounting and allocation layer of the deployed V2 architecture.

Exposed functionality

- Fee calculation from an AUD amount.
- Total protocol fee basis points.
- AETR allocation basis points.
- BTC allocation basis points.
- Cumulative fee accounting.
- AETR allocation tracking.
- BTC allocation tracking.

The fee router also exposes administrative controls for the fee collector and Treasury configuration.

10. Treasury and Cold-Wallet Architecture

The Treasury is designed as a separate reserve layer rather than treating active Vault collateral and protocol reserves as the same operational pool.

Treasury functions

- ETH balance reporting.
- ERC-20 token balance reporting.
- Designated cold-wallet identification.
- Controlled ETH withdrawals.
- Controlled token withdrawals.

The deployed Treasury contract identifies a coldWallet address and provides reserve accounting functions.

Cold-wallet custody reduces online exposure but does not create an absolute guarantee against compromise, loss or operational error.

11. AETR Token

AETR is the native protocol token of the AETHERIS ecosystem. The deployed V2 AETR contract exposes a maximum supply, allocation constants, controlled future-emission functionality and token burning.

Contract-level functionality

- MAX_SUPPLY
- INITIAL_SUPPLY

- Future emission accounting
- Emission controller
- Burn functionality

The public dashboard reads these values directly from the deployed contract so that tokenomics information can be compared with the on-chain source of truth.

12. AETR Tokenomics

The deployed AETR contract exposes allocation categories including Airdrop Allocation, Future Emission, Holder Reward Allocation, Treasury Allocation and Vault Incentive Allocation.

Category	Status
Airdrop Allocation	Contract-defined allocation
Future Emission	Contract-defined future emission allocation
Holder Reward Allocation	Contract-defined allocation
Treasury Allocation	Contract-defined allocation
Vault Incentive Allocation	Contract-defined allocation

Live supply, allocation and emission values should be read from the deployed AETR contract and the AETHERIS dashboard. This document does not substitute for on-chain verification.

13. Staking

The deployed Staking contract provides a basic AETR staking layer. It records individual balances and total staked AETR.

- stake(uint256 amount)
- withdraw(uint256 amount)
- balances(address)
- totalStaked()

The website provides an interface for monitoring and interacting with the deployed staking functionality.

14. Security Architecture

AETHERIS V2 uses multiple layers of controls. Security is treated as a system property rather than a single feature.

Access Control

Privileged functions are restricted by contract-level authorization.

Oracle Freshness

Price data includes timestamps and freshness controls.

Vault Risk Controls

Borrowing and liquidation are governed by explicit parameters.

Pause Controls

Relevant V2 contracts expose pause and unpause functionality.

Treasury Separation

Reserve custody is separated from active user collateral.

Role-Based AUDS

Minting and burning are protected by access-control roles.

Emission Control

Future AETR emission is controlled through an emission-controller mechanism.

Security disclaimer: No smart-contract protocol can honestly guarantee that it is impossible to hack. AETHERIS security should be evaluated through code review, testing, auditing, monitoring, operational controls and responsible disclosure.

15. Risk Management

A collateralized monetary system has several important risk categories. AETHERIS V2 addresses these through explicit contract parameters and operational separation.

- **Collateral Risk:** Collateral prices can change rapidly and affect Vault health.
- **Oracle Risk:** Incorrect or stale price data can affect protocol decisions.
- **Smart Contract Risk:** Code vulnerabilities can cause loss or incorrect execution.
- **Liquidation Risk:** Rapid market moves can create difficult liquidation conditions.
- **Liquidity Risk:** Market liquidity may affect the practical execution of asset transactions.
- **Operational Risk:** Administrative keys and custody procedures must be protected.
- **Governance / Configuration Risk:** Protocol parameters and privileged roles require careful management.

16. Ethereum Sepolia Testnet

The current public AETHERIS V2 deployment is integrated with Ethereum Sepolia. The website allows users to connect a compatible wallet and interact with the deployed testnet contracts.

Recommended test workflow

- Connect MetaMask to Ethereum Sepolia.
- Obtain Sepolia test ETH.
- Open the AETHERIS Dashboard.
- Deposit test ETH into the Vault.
- Test AUDS borrowing.
- Test AUDS repayment.
- Test withdrawals where the position permits them.
- Review Oracle, Treasury, Tokenomics and AUDS dashboards.

Sepolia is a test environment. Testnet assets do not represent production assets or protocol reserves.

17. Smart Contract Addresses

Contract	Sepolia Address
AETR Token	0xA6E6B409d1C40df1508bD06dC3B6f03f3CfeE66f
AUDS Stablecoin	0x614828e0b0db723e2B15196c6c6EcD230bf960A6
Price Oracle	0xe8a3b616fa79C77908F304AB7C0b03976295c4f0
Treasury	0xf8e361Ae009bEE83FB78bcD7B10Dbb4839413B40
Vault	0xF7DaA3b8DBFc3E923ce9645BA803d5Cff86d38C6
Protocol Fee Router	0x14830D7463C51c1EDf78f42bCC93D7017c306211
Staking	0x07f1752864abcFA1AE67742dF61E3ADD368f22b8

Network: Ethereum Sepolia

18. Roadmap

The roadmap below distinguishes the current V2 foundation from future protocol development.

Phase 1

V2 Sepolia deployment, Vault, AUDS, AETR, Oracle, Treasury, Staking and Protocol Fee Router.

Phase 2

Public testnet interface, documentation, tokenomics dashboard, risk monitoring and community testing.

Phase 3

Security review, testing expansion, monitoring infrastructure and preparation for broader testnet participation.

Phase 4

Future production-readiness work subject to audits, risk review, economic validation and governance decisions.

Future functionality should not be interpreted as deployed functionality until it is implemented, tested and independently verified on-chain.

19. Disclaimer

This whitepaper is an informational and technical description of the AETHERIS V2 protocol architecture. It is not financial, legal, tax or investment advice.

AETHERIS V2 is currently presented through Ethereum Sepolia testnet infrastructure. Testnet tokens have no intended production value. Smart-contract interactions involve technical and economic risks, including smart-contract vulnerabilities, oracle failures, market volatility, liquidation risk, liquidity risk, key-management risk and operational failures.

No statement in this document should be interpreted as a guarantee that AUSD will always trade at exactly USD 1, that the protocol cannot lose funds, or that the protocol is impossible to compromise.

Users should independently verify contract addresses, source code, transaction details and network information before interacting with any AETHERIS deployment.

AETHERIS Protocol

Ethereum Sepolia · AETHERIS V2

<https://github.com/AEtheris-01/aetheris-protocol>

https://x.com/Aetheris_pro